

Faster, cleaner way to extract lithium from battery waste

核心词汇

- **tremendous** adj. 极大的, 巨大的; 惊人的; 极好的
- **pollution** n. 污染
- **provide** v. 提供, 供应, 供给; 规定
- **contemporary** adj. 现代的, 当代的; 发生 [存在] 于同一时代的
- **threat** n. 威胁, 恐吓; 凶兆
- **harm** n. 伤害; 损害 v. 伤害; 危害; 损害
- **combine** v. 使化合; 使联合, 使结合 n. 联合收割机; 联合企业
- **consistent** adj. 一贯的, 一致的
- **considerable** adj. 相当大 (或多) 的, 可观的
- **ensure** v. 保证, 确保; 使安全
- **trade** n. 贸易, 交易; 行业; 职业 v. 交易, 买卖; 以物易物
- **threaten** v. 威胁; 恐吓; 预示
- **insight** n. 洞察力; 洞悉
- **implement** v. 实施, 执行; 实现, 使生效 n. 工具, 器具; 手段
- **aware** adj. 意识到的; 知道的; 有... 方面知识的; 懂世故的 n. (Aware) 人名; (阿拉伯、索) 阿瓦雷
- **attain** v. 完成; 获得; 达到
- **practice** n. 实践; 练习; 惯例 v. 练习; 实习; 实行
- **emit** v. 发出, 放射; 发行; 发表
- **oxide** n. [化学] 氧化物
- **regulation** n. 管理; 规则; 校准 adj. 规定的; 平常的
- **secure** adj. (from, against) 安全的 v. 得到; 使安全, 保护
- **metal** n. 金属, 金属制品
- **especially** adv. 尤其; 特别, 格外; 专门地, 主要地

- **origin** n. 起源；原点；出身；开端
- **prospective** adj. 预期的
- **promote** v. 促进,发扬；提升,提拔
- **acid** n. 酸,酸性物质 adj. 酸的,酸味的；尖刻的
- **moreover** adv. 再者，加之，而且

双语正文

Faster, Cleaner, and Targeted: A Microwave Fix for Lithium's White Gold Recycling Logjam

更快、更清洁、更精准：破解锂"白金"回收僵局的微波方案

Call it clean energy's bittersweet backbone: **lithium**, the silvery-white **metal** lauded as "white gold," powers everything from **contemporary** smartphones to grid-scale storage arrays that keep renewables humming when the sun hides or winds die down. But this critical resource faces a dual **threat**: a supply chain vulnerable to geopolitical **trade** tensions and environmental harm from primary extraction, paired with woeful recycling rates that waste **considerable** reserves locked in spent lithium-ion batteries (LIBs). A team at Rice University, led by Pulickel Ajayan, has unveiled a breakthrough solution that could defuse this tension—**combine** cutting-edge microwave heating with a biodegradable deep eutectic solvent (DES) to **attain** 50% lithium recovery in 30 seconds, leapfrogging a persistent industry bottleneck.

不妨称其为清洁能源苦乐参半的支柱：锂这种被誉为"白金"的银白色金属，为从当代智能手机到电网级储能阵列的各类设备供电，让可再生能源在阴天或无风时仍能稳定运转。但这一关键资源面临双重威胁：一是供应链易受地缘政治贸易紧张局势影响，二是原生开采带来的环境损害，再加上极低的回收率浪费了报废锂离子电池（LIBs）中封存的大量储备。莱斯大学由 Pulickel Ajayan 领导的团队推出了一项突破性解决方案，有望缓解这一困境——将前沿微波加

热与可生物降解的低共熔溶剂（DES）结合，可在 30 秒内实现 50% 的锂回收率，一举突破长期困扰行业的瓶颈。

First, let's confront the scale of the problem. The global LIB market, valued at over \$65 billion in 2023, is projected to surge by more than 23% over the next eight years, driven by **prospective** net-zero regulations and EV adoption. Though lithium is relatively abundant in Earth's crust, its primary **origin** sources—hard-rock mines in Australia and brine pools in Chile, Argentina, and Bolivia—**emit** massive volumes of greenhouse gases and **pollution**: each ton of mined lithium can require up to 500,000 gallons of water, destroy fragile ecosystems, and displace local communities. Recycling should be the obvious silver bullet, but conventional **practice** has fallen abysmally short. Traditional hydrometallurgy relies on corrosive **acid** leaching that produces toxic sludge, while more eco-friendly alternatives struggle to balance efficiency, cost, and **consistent** output. Even state-of-the-art facilities **secure** less than 5% of lithium from spent cathodes (often lithium cobalt nickel **oxide**), leaving most of it trapped or lost amid the recovery of higher-value metals like cobalt and nickel, which are prioritized because they precipitate first. **Moreover**, these processes take hours or days, guzzle energy, and degrade solvents over time.

首先，让我们正视这一问题的规模。受预期中的净零排放法规和电动汽车普及的推动，2023 年全球锂离子电池市场规模已超过 650 亿美元，预计未来 8 年将增长

23% 以上。尽管锂在地壳中相对丰富，但其主要来源——澳大利亚的硬岩矿以及智利、阿根廷和玻利维亚的盐池——会排放大量温室气体和污染：每开采 1 吨锂最多需要 50

万加仑的水，破坏脆弱的生态系统，并迫使当地社区搬迁。回收本应是显而易见的解决方案，但传统的做法却表现极差。传统湿法冶金依赖腐蚀性酸浸出工艺，会产生有毒污泥；而更环保的替代方案则难以兼顾效率、成本和稳定的产出。即使是最先进的回收厂，从报废阴极（通常是锂钴镍氧化物）中回收的锂也不足

5%，大部分锂在回收钴、镍等更高价值金属的过程中被截留或丢失——这些金属因优先沉淀而被优先回收。此外，这些工艺需要数小时甚至数天，消耗大

量能源，且溶剂会随时间降解。

Ajayan's team set out with a clear **insight**: what if they could flip the script, targeting lithium before other metals? To do this, they turned to a DES—a low-toxicity, readily biodegradable mixture—of choline chloride (a food additive) and ethylene glycol (a common antifreeze component), which they'd **aware** of from prior work showing it surrounds lithium with chloride ions and draws it into solution. Here's the clever microwave twist: only choline chloride efficiently absorbs microwave radiation, while ethylene glycol and the bulk metal **oxide** cathode remain largely inert at short bursts. Unlike an oil bath, which transfers heat slowly through conduction and convection, a kitchen-style microwave zaps energy directly to choline chloride molecules, heating the DES locally and supercharging the lithium-specific leaching reaction. In their Advanced Functional Materials study, they detail how this selective heating **ensures** that only lithium dissolves in 30 seconds (for 50% recovery) or 15 minutes (for an impressive 87% recovery)—**100 times faster** than oil bath methods, which also cause solvent decomposition over their 12-hour runs. Ajayan

的团队带着明确的洞察力开启了这项研究：如果他们能改变思路，先于其他金属靶向回收锂呢？为此，他们选用了一种由氯化胆碱（一种食品添加剂）和乙二醇（一种常见防冻液成分）组成的低毒性、易生物降解的低共熔溶剂（DES）——他们从先前的研究中了解到，这种溶剂会用氯离子包裹锂并将其溶解。这里的巧妙之处在于微波的应用：只有氯化胆碱能高效吸收微波辐射，而乙二醇和主体金属氧化物阴极在短时间微波照射下基本呈惰性。与通过传导和对流缓慢传热的油浴不同，厨房式微波炉能将能量直接传递给氯化胆碱分子，局部加热低共熔溶剂，加速针对锂的浸出反应。在他们发表于《Advanced Functional

Materials》的研究中，团队详细说明了这种选择性加热如何确保仅锂在 30 秒内溶解（回收率 50%），或在 15 分钟内实现可观的 87% 回收率——比油浴法快 100 倍，而油浴法在 12 小时的运行过程中还会导致溶剂分解。

This isn't just a lab curiosity. The Rice process could **provide** a path to **implement**

sustainable LIB recycling at scale, **promote** circularity in the clean energy sector, and reduce both environmental **harm** and reliance on volatile primary sources. "By prioritizing lithium and cutting energy use to a minimum, we're turning recycling into a practice that makes both economic and ecological sense," says Ajayan, Rice's Benjamin M. and Mary Greenwood Anderson Professor of Engineering. With governments worldwide drafting stricter **regulations** for LIB end-of-life, this breakthrough could not be timelier. (Word count: 762)

这并非只是实验室里的新奇发明。莱斯大学的这项工艺提供了一条规模化实施可持续锂离子电池回收的路径，能够促进清洁能源领域的循环发展，同时减少环境损害和对不稳定原生锂来源的依赖。莱斯大学工程学 Benjamin M. 和 Mary Greenwood Anderson 讲席教授 Ajayan

表示："通过优先回收锂并将能耗降至最低，我们正在将回收变成一种兼具经济和生态意义的实践。"随着全球各国政府为锂离子电池的报废处理制定更严格的法规，这一突破的出现恰逢其时。（词数：762）